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ORIGINAL ARTICLE

Impact of renal function on antidiabetic drug selection in Japan: A real-world analysis using the J-DREAMS database

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ABSTRACT

Aims: To describe renal function-specific prescribing patterns of antidiabetic drugs in Japanese patients with type 2 diabetes using real-world data and to identify selection trends up to the second therapy.

Materials and Methods: We retrospectively analyzed data from the J-DREAMS registry (2015–2023) to evaluate the proportions of first and second drug classes stratified by renal function (eGFR ≥60 vs <60 mL/min/1.73 m^2) and to examine temporal changes across registration periods. Prescription proportions by renal function were also compared using propensity score matching.

Results: Among 24,868 patients, 36.1% had an eGFR <60 mL/min/1.73 m^2. DPP-4 inhibitors were the most common first agents regardless of renal function, followed commonly by metformin and insulin. The use of SGLT2 inhibitors and GLP-1 receptor agonists increased as second-line therapies; in more recent periods, SGLT2 inhibitors were most frequently selected in the normal renal function group (25.6%), while GLP-1 receptor agonists ranked third in the reduced renal function group (15.9%). After matching, prescription proportions of SGLT2 inhibitors were similar between groups, whereas sulfonylureas (SUs), metformin, and thiazolidinediones were less frequent in the reduced renal function group.

Conclusions: Prescribing patterns in Japan have shifted toward earlier use of SGLT2 inhibitors and GLP-1 receptor agonists, with medications appropriately tailored to renal function.

INTRODUCTION

Currently, ten pharmacological classes are available for the treatment of type 2 diabetes mellitus (T2DM), and therapeutic strategies have evolved considerably over time. Initially, sulfonylureas (SUs), insulin preparations, and metformin were the mainstay treatments. In the 1990s, the introduction of alpha-glucosidase inhibitors (α-GIs), thiazolidinediones (TZDs), and glinides expanded the therapeutic arsenal. Subsequent studies demonstrated that achieving glycemic control reduces the risk of microvascular complications[1], and metformin was shown to improve long-term outcomes more effectively than SUs or insulin preparations[2]. Consequently, metformin became the first-line agent in many Western clinical guidelines.

In 2009, dipeptidyl peptidase-4 (DPP-4) inhibitors, which enhance insulin secretion by inhibiting incretin degradation, became available. In Japan, where patients typically exhibit lower intrinsic insulin secretory capacity and a high prevalence of elderly individuals and those with impaired renal function, DPP-4 inhibitors have been widely adopted as initial therapy due to their favorable safety profile[3]. Since around 2015, sodium–glucose cotransporter 2 (SGLT2) inhibitors and glucagon-like peptide-1 (GLP-1) receptor agonists have gained prominence, supported by growing evidence of their cardiovascular and renal protective effects. While metformin had long been recommended as the first-line treatment, recent guidelines

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from the American Diabetes Association (ADA) and the European Association for the Study of Diabetes (EASD) have increasingly emphasized individualized therapy based on cardiovascular and renal risk profiles[4].

Diabetic nephropathy, a major complication of diabetes, typically begins with albuminuria and progresses to a decline in estimated glomerular filtration rate (eGFR). Several antidiabetic agents, including metformin, require dose adjustment and become progressively contraindicated as renal impairment worsens, particularly when eGFR falls below 60 mL/min/1.73 m^2 due to concerns regarding efficacy and safety. As renal function declines, treatment options become increasingly limited, posing a significant clinical challenge. Although previous studies have examined the distribution of initial prescriptions and the usage rates of individual drug classes in Japan[3, 5–7], few have systematically analyzed how declining renal function influences drug selection patterns—particularly regarding the addition of second or later therapies—using large-scale database analyses.

Therefore, this study aimed to retrospectively investigate real-world prescribing patterns of antidiabetic agents according to renal function using J-DREAMS (Japan Diabetes compREhensive database project based on an Advanced electronic Medical record System), a nationwide, EMR-linked diabetes database, and to examine whether Japanese patients with type 2 diabetes and impaired renal function receive glycemic management comparable to those with preserved renal function.

MATERIALS AND METHODS

Study settings

J-DREAMS is a nationwide diabetes database jointly developed by the National Center for Global Health and Medicine (NCGM) and the Japan Diabetes Society (JDS)[8]. Data are collected from medical institutions across Japan, primarily large hospitals equipped with SS-MIX2 (Standardized Structured Medical record Information eXchange 2) systems. Clinical data are extracted from the Standard Diabetes Management Template (SDMT), which is integrated into the electronic medical record systems at participating institutions.

The SDMT captures detailed patient-level information, including medical history, diabetes-related complications and comorbidities, body weight, and blood pressure. These data, along with basic demographic information, laboratory test results (blood and urine), and prescription records, are automatically transmitted and stored on a centralized server. The diagnosis of diabetes mellitus is made according to the criteria established by the Japan Diabetes Society.

Patient selection

We included patients registered in the J-DREAMS database between December 1, 2015, and March 31, 2023, who met the following inclusion criteria: (1) a documented diagnosis of T2DM, and (2) availability of follow-up data for at least 6 months after the initial record. Exclusion criteria were: (1) age under 18 years at the start of observation, and (2) a recorded diagnosis of type 1 diabetes mellitus or gestational diabetes mellitus.

The index date (start of observation) was defined as the date with the highest serum creatinine value measured within a ±90-day window surrounding the date of registration. Patient background data—including age, sex, ethnicity, body mass index (BMI), smoking status, duration of diabetes, hemoglobin A1c (HbA1c), systolic and diastolic blood pressure, and eGFR—were extracted for all eligible patients.

Hypertension was defined as a systolic blood pressure ≥140 mmHg and/or a diastolic blood pressure ≥90 mmHg, or the prescription of antihypertensive medications. Dyslipidemia was defined as low-density lipoprotein cholesterol (LDL-C) ≥120 mg/dL and/or the prescription of lipid-lowering agents. Chronic kidney disease (CKD) was classified according to the Kidney Disease: Improving Global Outcomes (KDIGO) guidelines. Patients with CKD Stage G1–2 and albuminuria category A1 were considered free of diabetic nephropathy[9].

Diabetic retinopathy was diagnosed by ophthalmologists, with the final diagnosis confirmed by the attending physician based on shared clinical information. Diagnoses of diabetic neuropathy, coronary artery disease, cerebrovascular disease, peripheral artery disease, chronic heart failure, and malignancy were made in accordance with relevant clinical guidelines or at the discretion of the attending physicians.

Primary outcome

The primary outcome was the proportion of each antidiabetic drug class used as first and second therapy, stratified by renal function. Patients were categorized into two groups based on their eGFR: the normal renal function group (eGFR ≥60 mL/min/1.73 m^2; CKD stages G1–G2) and the reduced renal function group (eGFR <60 mL/min/1.73 m^2; CKD stages G3a–G5).

The first drug class was defined as follows: for drug-naïve patients at the time of J-DREAMS registration, the first antidiabetic drug class prescribed after registration was considered the first agent. For patients already receiving antidiabetic therapy at registration, the drug class prescribed at that time was defined as the first agent. The second drug class was defined as any new class added after the initial agent, regardless of whether the patient was drug-naïve or already on treatment at registration (Figure S1). We also evaluated the time to the addition of the next antidiabetic class according to renal function, as well as the HbA1c level at the time of addition.

Additionally, patients were divided into two subgroups based on the registration period: those registered between December 1, 2015, and April 30, 2020, and those registered on or after May 1, 2020. The same analysis was conducted within each subgroup.

Antidiabetic drug classes were categorized as follows: biguanides (BGs), SUs, rapid-acting insulin secretagogues (glinides), DPP-4 inhibitors, GLP-1 receptor agonists, SGLT2 inhibitors, α-GIs, TZDs, and insulin preparations. Imeglimin was excluded from classification, as it was not available for most of the study period.

Secondary outcome

The secondary outcome was the comparison of prescription ratios for each antidiabetic drug class between the normal and reduced renal function groups, using propensity score matching. Propensity scores were estimated using logistic regression models with age, sex, BMI, and HbA1c as covariates. A 1:1 nearest-neighbor matching algorithm without replacement was applied to balance baseline characteristics between the two groups. In addition, disease duration, eGFR, and the prevalence of complications were also calculated according to renal function, both before and after matching.

Statistical analysis

Data from eligible patients were extracted from the J-DREAMS database. For continuous variables, the number of observations, means, and standard deviations (SDs) were calculated. For categorical and ordinal variables, frequencies and percentages were summarized. Comparisons of prescription ratios by renal function group for each antidiabetic drug class were performed using the chi-squared test or Fisher's exact test, as appropriate.

For the secondary outcome, baseline characteristics were compared between the normal and reduced renal function groups using Welch's *t*-test for continuous variables and the chi-squared test or Fisher's exact test for categorical variables. A two-sided *P*-value of <0.05 was considered statistically significant. All statistical analyses were conducted using STATA/MP versions 17 and 18 (StataCorp LLC, College Station, TX, USA).

RESULTS

Study population

Between December 1, 2015, and March 31, 2023, a total of 87,781 individuals with diabetes were registered in the J-DREAMS database (Figure S2). Out of these, 24,868 patients met the eligibility criteria and were included in the present analysis. The mean age of the study population was 66.0 ± 12.0 years, and 38.3% were female. All participants were of Japanese ethnicity (Table 1).

The mean duration since the diagnosis of type 2 diabetes was 13.7 ± 10.8 years. Baseline clinical characteristics included a mean hemoglobin A1c (HbA1c) of 7.30 ± 1.20%, a mean BMI of 25.5 ± 5.0 kg/m^2, and a mean eGFR of 71.5 ± 25.2 mL/min/1.73 m^2.

Renal function and other comorbidities/complications

Among the study population, 36.1% had an eGFR <60 mL/min/1.73 m^2, including 5.4% with an eGFR <30 mL/min/1.73 m^2. The proportions of patients with a urinary albumin-to-creatinine ratio (ACR) between 30 and 299 mg/gCr and those with ACR ≥300 mg/gCr were both 15.2%. Overall, 32.1% of patients had either an ACR ≥30 mg/gCr or an eGFR ≤30 mL/min/1.73 m^2, corresponding to stage 2 or later diabetic kidney disease according to the Japanese classification.

Table 1 | Clinical and laboratory features, comorbid conditions/complications of the patients

	Total, *n*	*n* (%)	Average (SD)
Age (years)	24,868		66.0 (12.0)
Sex (Female)	24,868	9,513 (38.3)	
Race (Japanese)	24,868	24,868 (100)	
BMI (kg/m^2)	22,765		25.5 (5.0)
Smoking (nonsmoker)	15,144	6,679 (44.1)	
Duration of diabetes (years)	16,620		13.7 (10.8)
HbA1c (%)	24,868		7.30 (1.20)
Systolic blood pressure (mmHg)	21,699		129.7 (15.8)
Diastolic blood pressure (mmHg)	21,643		73.2 (11.6)
eGFR (mL/min/1.73 m^2)	24,868		71.5 (25.2)
Hypertension	24,868	10,257 (41.3)	
Dyslipidemia	24,868	10,159 (40.9)	
Diabetic neuropathy	24,868	3,544 (14.3)	
Diabetic retinopathy	24,868	4,547 (18.3)	
Diabetic nephropathy stage 2 and higher	23,237	7,450 (32.1)	
Category of CKD	23,237		
G1 (≧90)		3,634 (14.6)	
G2 (60–89)		12,195 (49.1)	
G3a (45–59)		5,255 (21.1)	
G3b (30–44)		2,442 (9.8)	
G4 (15–30)		947 (3.8)	
G5 (<15)		395 (1.6)	
A1		16,052 (69.6)	
A2		3,495 (15.2)	
A3		3,511 (15.2)	
Coronary artery disease	24,868	2,257 (9.1)	
Cerebrovascular disease	24,868	1,280 (5.2)	
Peripheral artery disease	24,868	558 (2.2)	
Chronic heart failure	24,868	766 (3.1)	
Amputation	24,868	102 (0.4)	
Malignancy	24,868	3,228 (13.0)	

BMI, body mass index; CKD, chronic kidney disease; eGFR, estimated glomerular filtration rate; HbA1c, hemoglobin A1c.

Hypertension was present in 41.3% of patients, and dyslipidemia in 40.9%. Microvascular complications included diabetic retinopathy in 18.3% and diabetic neuropathy in 14.3%. The prevalence of macrovascular and other complications was as follows: coronary artery disease, 9.1%; cerebrovascular disease, 5.2%; peripheral artery disease, 2.2%; congestive heart failure, 3.1%; and malignancies, 13.0%.

Primary outcome

For the first therapy, which was defined as the initial antidiabetic drug class used at or after registration, the largest proportions in the normal renal function group (eGFR ≥60 mL/min/1.73 m^2) were DPP-4 inhibitors (32.6%), BGs (29.0%), insulin

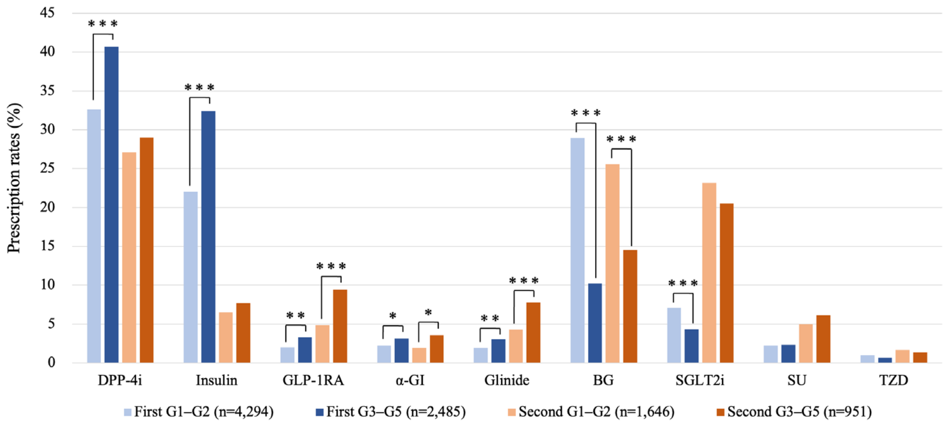

Figure 1 | Class-specific prescription rates by renal function for first and second antidiabetic therapies. This figure shows the proportions of each antidiabetic drug class selected as first and second therapies in patients with normal renal function (G1–G2, eGFR ≥60 mL/min/1.73 m^2) and those with reduced renal function (G3–G5, eGFR <60 mL/min/1.73 m^2). The four groups defined by treatment line and renal function category are indicated in the figure legend. Statistical significance between renal function groups for each drug class is shown as follows: *$P < 0.05$, **$P < 0.01$, and ***$P < 0.001$. BG, biguanide; DPP-4i, dipeptidyl peptidase-4 inhibitor; GLP-1RA, glucagon-like peptide-1 receptor agonist; SGLT2i, sodium–glucose cotransporter 2 inhibitor; SU, sulfonylurea; TZD, thiazolidinedione; α-GI, α-glucosidase inhibitor.

preparations (22.0%), and SGLT2 inhibitors (7.1%) (Figure 1, Table 2). In contrast, among patients with reduced renal function (eGFR <60 mL/min/1.73 m^2), DPP-4 inhibitors (40.7%) and insulin preparations (32.4%) were most commonly used, followed by BGs (10.2%) and SGLT2 inhibitors (4.3%). Patients with reduced renal function had significantly higher prescription ratios of DPP-4 inhibitors and insulin preparations than those with normal renal function. Although the overall use of GLP-1 receptor agonists, α-GIs, and glinides was relatively low, these agents were significantly more frequently prescribed in the reduced renal function group. In contrast, BGs and SGLT2 inhibitors were significantly more commonly used in patients with normal renal function.

For the second therapy, which was defined as any new antidiabetic drug class subsequently added, the four leading drug classes in the normal renal function group were DPP-4 inhibitors (27.1%), BGs (25.6%), SGLT2 inhibitors (23.1%), and insulin preparations (6.5%). In the reduced renal function group, the four leading drug classes were DPP-4 inhibitors (29.0%), SGLT2 inhibitors (20.5%), BGs (14.5%), and GLP-1 receptor agonists (9.5%). Compared to the first therapy, both groups tended to have higher use of SGLT2 inhibitors and GLP-1 receptor agonists as second agents. GLP-1 receptor agonists, α-GIs, and glinides were significantly more frequently prescribed in the reduced renal function group, whereas BGs remained significantly more common in the normal renal function group,

Table 2 | Top four antidiabetic drug classes prescribed as first and second therapy by renal function and period

Period	Sequence of therapy	Renal group	1st most prescribed (%)	2nd most prescribed (%)	3rd most prescribed (%)	4th most prescribed (%)
Entire period	1st	G1–G2	DPP-4i (32.6)	BG (29.0)	Insulin (22.0)	SGLT2i (7.1)
		G3–G5	DPP-4i (40.7)	Insulin (32.4)	BG (10.2)	SGLT2i (4.3)
	2nd	G1–G2	DPP-4i (27.1)	BG (25.6)	SGLT2i (23.1)	Insulin (6.5)
		G3–G5	DPP-4i (29.0)	SGLT2i (20.5)	BG (14.5)	GLP-1RA (9.5)
Dec 2015 – Apr 2020	1st	G1–G2	DPP-4i (35.7)	BG (29.3)	Insulin (20.0)	SGLT2i (5.2)
		G3–G5	DPP-4i (44.4)	Insulin (30.7)	BG (10.0)	α-GI (3.5)
	2nd	G1–G2	DPP-4i (28.8)	BG (26.2)	SGLT2i (22.4)	Insulin (6.0)
		G3–G5	DPP-4i (30.6)	SGLT2i (20.4)	BG (14.5)	Glinide (8.1)
May 2020 – Mar 2023	1st	G1–G2	BG (28.2)	Insulin (27.4)	DPP-4i (24.2)	SGLT2i (12.0)
		G3–G5	Insulin (37.1)	DPP-4i (31.8)	BG (10.7)	SGLT2i (8.5)
	2nd	G1–G2	SGLT2i (25.6)	BG (23.5)	DPP-4i (21.7)	GLP-1RA (10.2)
		G3–G5	DPP-4i (24.0)	SGLT2i (20.6)	GLP-1RA (15.9)	BG (14.6)

BG, biguanide; DPP-4i, dipeptidyl peptidase-4 inhibitor; GLP-1RA, glucagon-like peptide-1 receptor agonist; SGLT2i, sodium–glucose cotransporter 2 inhibitor; α-GI, α-glucosidase inhibitor.

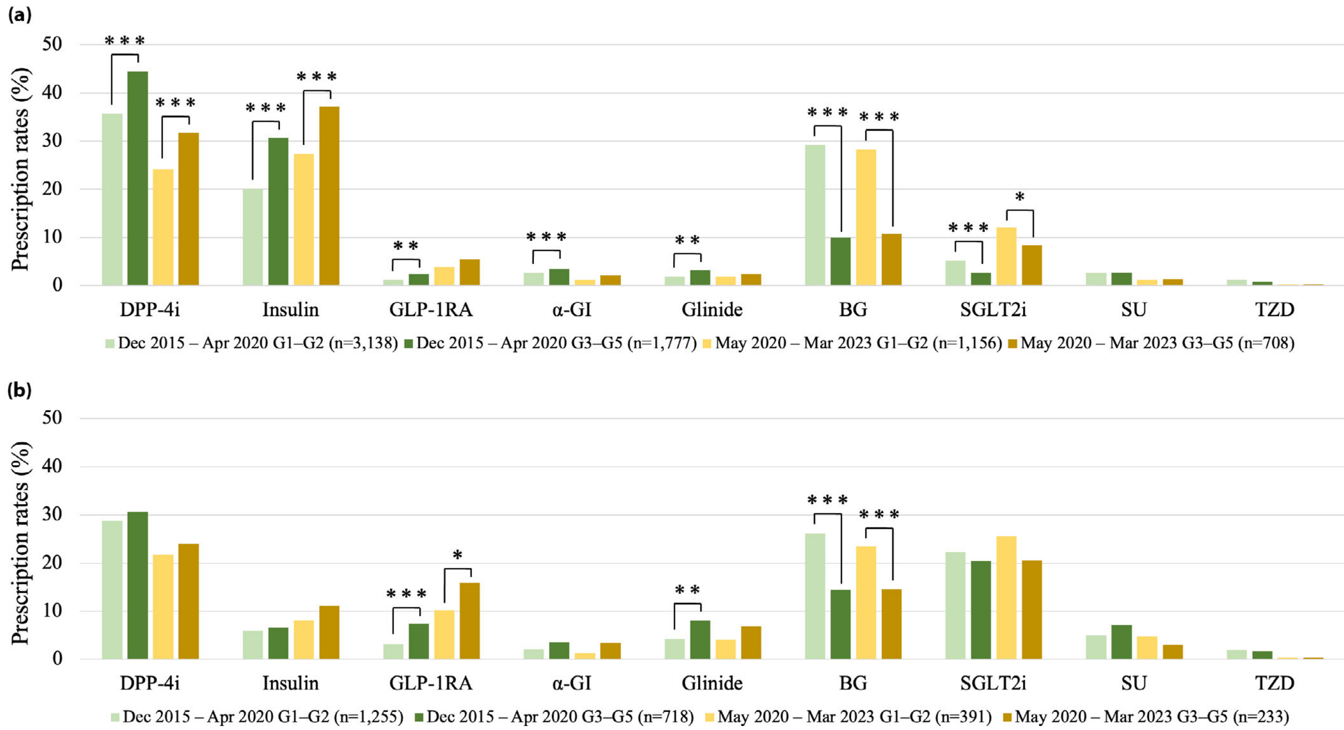

Figure 2 | Time trends in antidiabetic drug selection by renal function and treatment line. This figure illustrates temporal changes in the proportions of antidiabetic drug classes selected as first therapy (a) and second therapy (b), stratified by renal function and registration period. Two time periods were compared—December 2015 to April 2020 and May 2020 to March 2023—across four groups defined by renal function: patients with normal renal function (G1–G2, eGFR ≥60 mL/min/1.73 m^2) and those with reduced renal function (G3–G5, eGFR <60 mL/min/1.73 m^2). Statistical significance between renal function groups for each drug class is shown as follows: *P < 0.05, **P < 0.01, and ***P < 0.001. BG, biguanide; DPP-4i, dipeptidyl peptidase-4 inhibitor; GLP-1RA, glucagon-like peptide-1 receptor agonist; SGLT2i, sodium–glucose cotransporter 2 inhibitor; SU, sulfonylurea; TZD, thiazolidinedione; α-GI, α-glucosidase inhibitor.

and no significant difference was observed for SGLT2 inhibitors.

A time-stratified analysis using May 2020 as the cutoff revealed that the prescription rate of BGs remained relatively stable for both first and second therapies (Figure 2, Table 2). However, the use of DPP-4 inhibitors declined, while the relative use of SGLT2 inhibitors, GLP-1 receptor agonists, and insulin preparations increased. After May 2020, SGLT2 inhibitors (25.6%) became the most frequently prescribed second agent in the normal renal function group, while GLP-1 receptor agonists (15.9%) ranked third in the reduced renal function group.

As shown in Table S1, HbA1c increased significantly before treatment intensification in all groups, except for drug-naïve patients moving from first to second therapy, in whom HbA1c did not significantly change. The duration to treatment intensification is also provided in Table S1.

Secondary outcome

At baseline, patients in the normal renal function group were significantly younger (mean age: 63.0 ± 13.0 years vs 71.5 ± 10.5 years, P < 0.0001), more likely to be male (40.1% vs 34.9%, P < 0.001), and had a higher BMI (25.7 ± 5.2 vs 25.2 ± 4.4 kg/m^2, P < 0.0001) and HbA1c (7.35 ± 1.3% vs 7.22 ± 1.1%, P < 0.0001) (Table S2). The duration of diabetes was shorter in the normal renal function group (12.1 ± 10.0 years vs 16.7 ± 11.5 years, P < 0.0001), and a marked difference in eGFR was observed (80.5 ± 16.7 vs 44.3 ± 13.0 mL/min/1.73 m^2, P < 0.0001).

After 1:1 propensity score matching using age, sex, BMI, and HbA1c as covariates, the difference in diabetes duration remained statistically significant but was attenuated (14.3 ± 10.8 years vs 16.1 ± 11.0 years, P < 0.0001). Similarly, the difference in eGFR was slightly reduced but remained significant (77.0 ± 14.1 vs 44.6 ± 13.1 mL/min/1.73 m^2, P < 0.0001). Moreover, despite adjustment, the prevalence of microvascular and macrovascular complications remained significantly higher in the reduced renal function group.

In the pre-matching analysis, the prescription ratios of DPP-4 inhibitors, insulin preparations, α-GIs, glinides, and GLP-1 receptor agonists were significantly higher in the reduced renal function group (Figure 3). Conversely, BGs, SGLT2 inhibitors,

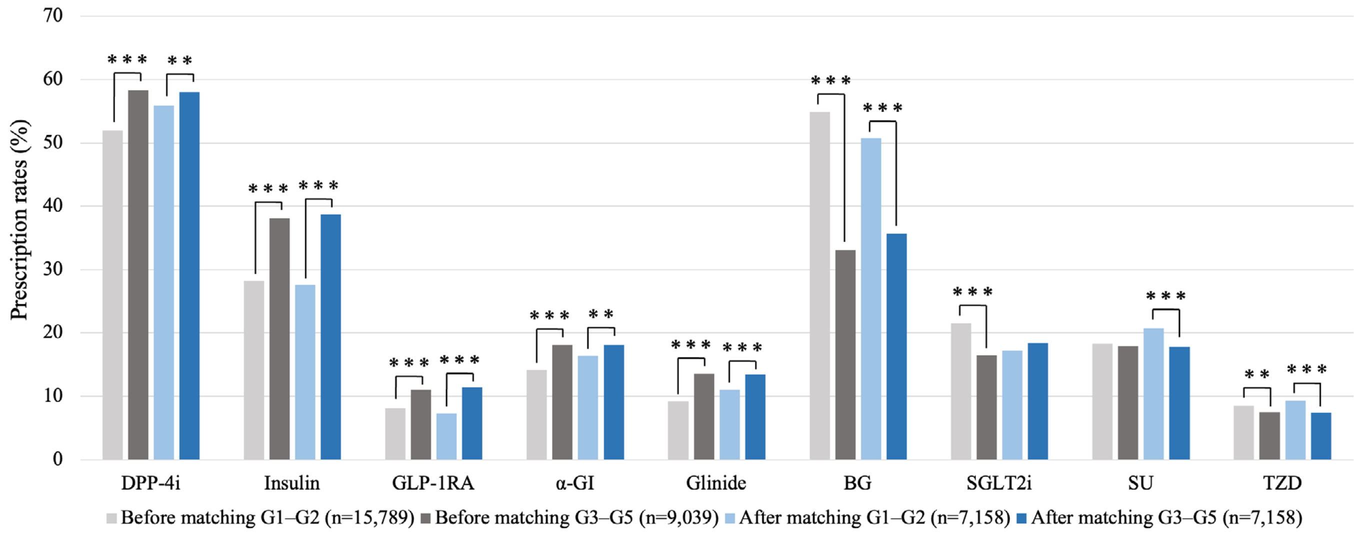

Figure 3 | Prescription rates by drug class before and after matching between renal function groups. This figure compares the proportions of each antidiabetic drug class used in patients with normal renal function (G1–G2, eGFR ≥60 mL/min/1.73 m^2) and those with reduced renal function (G3–G5, eGFR <60 mL/min/1.73 m^2) before and after propensity score matching. Before matching, the sample sizes were G1–G2 (n = 15,789) and G3–G5 (n = 9,039), whereas after matching both groups had n = 7,158. Statistical significance between renal function groups for each drug class is shown as follows: *P < 0.05, **P < 0.01, and ***P < 0.001. BG, biguanide; DPP-4i, dipeptidyl peptidase-4 inhibitor; GLP-1RA, glucagon-like peptide-1 receptor agonist; SGLT2i, sodium–glucose cotransporter 2 inhibitor; SU, sulfonylurea; TZD, thiazolidinedione; α-GI, α-glucosidase inhibitor.

and TZDs were more frequently prescribed in the normal renal function group. After propensity score matching, the increased use of DPP-4 inhibitors, insulin preparations, α-GIs, glinides, and GLP-1 receptor agonists in the reduced renal function group remained consistent. However, the difference in the use of SGLT2 inhibitors' use was no longer statistically significant. Additionally, SUs were significantly less frequently prescribed in the reduced renal function group.

DISCUSSION

In this study, we retrospectively analyzed real-world prescribing patterns of glucose-lowering agents in patients with T2DM according to renal function, using the J-DREAMS database—a nationwide, EMR-linked diabetes registry in Japan. To our knowledge, this is one of the first large-scale studies in Japan to systematically examine not only the first but also the second and subsequent prescribing patterns of antidiabetic agents according to renal function. Our findings indicate that antidiabetic medications were generally selected appropriately based on renal function in routine clinical practice, and recent shifts in prescribing patterns reflect the accumulation of clinical evidence over time.

Although several studies have reported on prescribing trends of antidiabetic agents in Japan, few have systematically examined drug selection with a specific focus on renal function, including both first and second-line therapies. Japanese treatment guidelines are characterized by their flexibility, allowing clinicians to select any drug class as first therapy based on individual patient pathophysiology[10]. Consistent with previous reports[5, 7], our study confirmed the high prescription ratios of DPP-4 inhibitors as initial therapy, reflecting their suitability for the Japanese T2DM phenotype. At the same time, our findings also demonstrated that patients with reduced renal function were more frequently treated with DPP-4 inhibitors and insulin, suggesting that therapeutic options become restricted when renal function declines.

The J-Discover study (2014–2015 data), a prospective observational cohort involving 1,806 Japanese patients with T2DM, reported that DPP-4 inhibitors, BGs, SUs, and α-GIs were the most commonly prescribed first therapies, while DPP-4 inhibitors, BGs, SGLT2 inhibitors, and SUs were frequently selected as second agents[6]. Compared to J-Discover, our study revealed notably higher usage ratios of insulin preparations and GLP-1 receptor agonists, likely due to the inclusion of data from specialized medical institutions. The increased use of SGLT2 inhibitors and GLP-1 receptor agonists as second agents is particularly noteworthy.

International comparisons further contextualize our findings. A retrospective cohort study using the UK Clinical Practice Research Datalink (CPRD) found that metformin was the most commonly prescribed agent regardless of renal function[11]. However, SGLT2 inhibitors were prescribed approximately three times more often in patients with normal renal function than in those with impaired renal function (28.5% vs 9.4%) as first therapy, and nearly six times more frequently as second therapy (46.3% vs 7.9%). In the United States, a cross-sectional analysis using the National Health and Nutrition Examination Survey (NHANES) reported a high prevalence of metformin use, while the combined prescription rate of SGLT2 inhibitors and GLP-1 receptor agonists was only 7.1%[12]. In contrast, our study demonstrated higher usage ratios of GLP-1 receptor agonists and early adoption of SGLT2 inhibitors even among patients with renal impairment, suggesting a more proactive implementation of emerging evidence in Japanese clinical practice.

In recent years, increasing attention has been paid to the cardiorenal protective effects of SGLT2 inhibitors and GLP-1 receptor agonists. SGLT2 inhibitors have demonstrated reductions in cardiovascular and renal events in the EMPA-REG OUTCOME trial and prevention of diabetic kidney disease (DKD) progression in the CREDENCE trial[13, 14]. Further evidence from the DAPA-CKD and EMPA-KIDNEY trials confirmed renoprotective effects even in patients without diabetes, establishing SGLT2 inhibitors as a cornerstone therapy for CKD[15,16]. Similarly, GLP-1 receptor agonists have shown potential renoprotective effects in the LEADER, SUSTAIN-6, and REWIND trials, with more definitive evidence provided by the recent FLOW trial[17–20]. Currently, SGLT2 inhibitors and GLP-1 receptor agonists, alongside renin-angiotensin system (RAS) inhibitors and mineralocorticoid receptor antagonists (MRAs), are considered part of the "four pillars" of early DKD management[21]. Japanese patients with T2DM have been reported to experience higher ratios of heart failure and CKD events compared to those in Europe and the United States. These epidemiological differences may partly explain the greater emphasis on agents with established cardiorenal protective effects in Japanese clinical practice[22].

In our study, 83.1% of patients in the reduced renal function group were classified as CKD Stage G3, indicating that many remained eligible for SGLT2 inhibitor therapy and were also likely to derive cardiorenal benefits from these agents. Although initial analyses showed lower SGLT2 inhibitor prescription ratios in this group, the difference disappeared after propensity score matching, suggesting appropriate use of SGLT2 inhibitors in patients with renal impairment. GLP-1 receptor agonists, which require minimal dose adjustment based on renal function, were significantly more frequently prescribed in the reduced renal function group, further supporting alignment with current guidelines and evidence.

Regarding medications requiring dose adjustment in renal impairment, metformin and SUs are associated with increased risks of lactic acidosis and hypoglycemia, respectively[23–25]. In our study, both agents were prescribed less frequently in the reduced renal function group, indicating cautious and appropriate use. Notably, while no significant difference in SU use was observed before matching, the prescription rate declined significantly in the reduced renal function group after matching, reflecting careful clinical decision-making. Although the prescription ratios declined, a certain proportion remained, which may reflect that many patients with reduced renal function in our cohort still had relatively preserved kidney function.

The clinical significance of this study lies in its large-scale, detailed analysis of contemporary prescribing patterns stratified by renal function in Japan. As Japanese treatment algorithms are often shaped by real-world practice, our findings provide valuable evidence to inform future renal function–based treatment protocols. In an era of increasing therapeutic complexity, there is a pressing need to establish practical, evidence-based algorithms that incorporate renal function as a key consideration.

This study has several limitations. First, as a retrospective observational study using a database, we could not directly assess physicians' intentions, patients' preferences, or medication adherence. Second, prescription records in the J-DREAMS database may be incomplete, and initial prescriptions may not always be accurately identified. Third, intra-class variations in dosage and formulation, drug interactions, and adverse events were not captured. Fourth, we could not fully exclude the possibility that SGLT2 inhibitors were prescribed primarily for non-diabetic indications such as CKD or heart failure. In particular, since J-DREAMS is a database of patients with diabetes, SGLT2 inhibitors may have been added as a second therapy to manage cardiovascular or renal risks even when glycemic control remained stable with the first agent. This clinical practice pattern, driven by cardiorenal protective evidence rather than insufficient glycemic control, could influence the interpretation of second therapy intensification trends. Fifth, although the reduced renal function group largely comprised patients with relatively preserved renal function (Stage G3), further investigation is needed in those with more advanced impairment (eGFR <30 mL/min/1.73 m^2), where therapeutic options are more restricted. Sixth, newer agents such as imeglimin, which can be used in patients with impaired renal function, were excluded from the present analysis because they were unavailable for much of the study period[26]. Finally, as the study was conducted mainly at specialized institutions, prescribing patterns in primary care settings may differ due to differences in patient characteristics, comorbidity burden, and clinical practice structures, and thus caution is warranted when generalizing these findings.

CONCLUSIONS

This large-scale, real-world study using the nationwide J-DREAMS database provides a comprehensive overview of contemporary prescribing patterns of antidiabetic medications in Japanese patients with T2DM, stratified by renal function. Our findings indicate that therapeutic choices were generally appropriate and aligned with renal function, with cautious use of agents such as metformin and SUs in patients with impaired renal function. The increasing use of SGLT2 inhibitors and GLP-1 receptor agonists over time reflects the proactive integration of emerging clinical evidence into routine practice.

Importantly, the study highlights the potential of real-world data to inform individualized treatment strategies and guide the development of renal function-based therapeutic algorithms tailored to the Japanese clinical setting. Further research is warranted to assess the long-term clinical outcomes associated with these prescribing patterns and to explore the real-world use and effectiveness of newer agents, including imeglimin and other emerging therapies.

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DISLOSURE

M.K. declares no conflicts of interest. M.O. reports receiving honoraria for lectures from Astellas Pharma Inc., MSD K.K., Otsuka Pharmaceutical Co., Ltd., Kissei Pharmaceutical Co., Ltd., Teijin Pharma Ltd., Nippon Boehringer Ingelheim Co., Ltd., Sanofi K.K., Abbott Japan LLC, Sumitomo Pharma Co., Ltd., Novo Nordisk Pharma Ltd., Kowa Co., Ltd., AstraZeneca K.K., Taisho Pharmaceutical Co., Ltd., Daiichi Sankyo Co., Ltd., Tanabe Pharma Corporation, Ono Pharmaceutical Co., Ltd., Eli Lilly Japan K.K., Takeda Pharmaceutical Co., Ltd., Novartis Pharma K.K., and Bayer Yakuhin Ltd.; receiving clinical commissioned/joint research grants from Novo Nordisk Pharma Ltd., Eli Lilly Japan K.K., Nippon Boehringer Ingelheim Co., Ltd., Sanofi K.K., Abbott Japan LLC, MSD K.K., Kyowa Kirin Co., Ltd., Astellas Pharma Inc., Sumitomo Pharma Co., Ltd., and Bayer Yakuhin Ltd., and scholarship grants from Sumitomo Pharma Co., Ltd. and Nippon Boehringer Ingelheim Co., Ltd. K.U. reports receiving honoraria for lectures from Ono Pharmaceutical Co., Ltd., Sanofi K.K., Sumitomo Pharma Co., Ltd., Nippon Boehringer Ingelheim Co., Ltd., Taisho Pharmaceutical Co., Ltd., Teijin Pharma Ltd., Bayer Yakuhin Ltd., Otsuka Pharmaceutical Co., Ltd., Abbott Japan LLC, Kowa Co., Ltd., Eli Lilly Japan K.K., Tanabe Pharma Corporation, and Novo Nordisk Pharma Ltd.; receiving grants from Sumitomo Pharma Co., Ltd.; and serving as Chair of the Board of Directors of the Japan Diabetes Society.

Approval of the research protocol: The study protocol was reviewed and approved by the Institutional Review Board of the National Center for Global Health and Medicine (NCGM; approval number: NCGM-S-004817-00) on March 8, 2024.

Informed consent: In accordance with the Guidelines for Epidemiological Studies issued by the Ministry of Health, Labour and Welfare of Japan, written informed consent was not required. Information regarding the J-DREAMS project was made publicly available on a dedicated website, and patients retained the right to withdraw their data from the registry at any time.

Approval date of registry and the registration no. of the study/trial: N/A.

Animal studies: N/A.

AUTHOR CONTRIBUTIONS

M.O. and K.U. designed the study and developed the study protocol. M.K. and M.O. analyzed the data. M.K. wrote the original draft and all authors contributed to the interpretation of the data and critically reviewed/revised the manuscript. All authors approved the final manuscript.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from J-DREAMS. Restrictions apply to the availability of these data, which were used under license for this study. Data are available from the author(s) with the permission of J-DREAMS.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of the article.

Figure S1 Data extraction flow. This flowchart illustrates how drug-naïve patients were assigned to first therapy and how patients already treated with one agent were classified into second therapy, followed by stratification into CKD stages G1–G2 and G3–G5.

Figure S2 Patient disposition. This flow diagram shows the selection process for the study cohort, including exclusions and the final number of patients included (n = 24,868) from the J-DREAMS database.

Table S1 Time to Next Antidiabetic Therapy and Changes in HbA1c According to Renal Function and Treatment Pattern.

Table S2 | Baseline characteristics before and after propensity score matching by renal function group.